Diagnosing Index-Shift and Convergence-Condition Errors

Find the mistake, compute the correct value, and name what went wrong.

Directions:

- Each problem shows one student's work with exactly one thing wrong in it.
- Part (a) asks for the correct value. Part (b) asks you to name the error in a sentence or two.
- Give exact values — fractions, not decimals — unless a problem says otherwise.

1. A student evaluates $\sum_{k=2}^{6} k$ by using the formula $\frac{n(n+1)}{2}$ with $n = 6$ and writes 21.

(a) Find the correct value of the sum. _________

(b) Name the error in the student's method.

2. A student rewrites $\sum_{n=1}^{4} 3^n$ as $\sum_{m=0}^{4} 3^{m+1}$ and evaluates the new sum as 363.

(a) Find the correct value of $\sum_{n=1}^{4} 3^n$. _________

(b) Name the error in the rewriting.

3. A student applies $\dfrac{a}{1-r}$ to $\sum_{n=0}^{\infty} \left(\frac{3}{2}\right)^n$ and reports a sum of -2.

(a) Evaluate $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$, which the formula *does* apply to. _________

(b) Explain why the same formula cannot be used on the student's series.

4. A student says $\lim\limits_{n\to\infty} \dfrac{3n+2}{n-5} = 1$, reasoning that "the 2 and the -5 disappear and the n's cancel".

(a) Find the correct limit. _________

(b) Name the error in the reasoning.

5. To rewrite $\sum\limits_{k=3}^{7}(k-2)^2$ with a new index $j = k-2$, a student writes $\sum\limits_{j=1}^{7} j^2$ and gets 140.

(a) Find the correct value of $\sum\limits_{k=3}^{7}(k-2)^2$. _________

(b) Name the error, and say how many terms the sum should contain.

6. A student evaluates $\sum_{n=1}^{\infty} 5\left(\frac{1}{4}\right)^{n-1}$ as $\dfrac{5/4}{1-1/4} \approx 1.67$.

(a) Find the correct sum as an exact fraction. __________

(b) Name the error in the value used for a.

7. A student claims the sequence $a_n = (-1)^n \dfrac{n}{n+1}$ converges to 1, because $\dfrac{n}{n+1} \to 1$.

(a) Compute a_{10} and a_{11} as exact fractions. $a_{10} =$ __________ $a_{11} =$ __________

(b) Use those two values to name the error in the student's claim.

8. A student evaluates $\sum_{n=2}^{\infty} 3\left(\frac{1}{2}\right)^n$ as $\dfrac{3}{1-1/2} = 6$.

(a) Find the correct sum as an exact fraction. _________

(b) Name the error, and state what a should have been.